

Alexandre Baron

3D Graphics & Animation Programmer

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I'm passionate about programming game engines and 3D simulations, with more than a decade of experience in the video game industry. I'm always eager to know more about the rendering technologies behind the products of the future. I also have a passion to share the knowledge I gathered during my journey through teaching.

Professional Experience

3Cs Programming Team Lead (Assassin's Creed Hexe, ongoing)

Ubisoft, Montréal, QC, Canada

- Management of the 3Cs programming team (Character, Controller, Camera) on Assassin's Creed Hexe.
- Supervising collaboration with the technical animation team to implement new animation features:
- Motored ragdolls, environment-aware IK, inertialization (adjustment blending).
- Programming support in C++ with Anvil Engine.
- Technologies: Anvil, C++, **Perforce**, **Jira**, Confluence

 **UBISOFT**
MONTREAL
July 2024–today

Programming Team Lead (MOD Snowdrop)

Ubisoft

January–June 2024

- Leading a training game project for new employees joining the company (5 programmers).
- Repurposing existing tech to make a game released internally after 6 months.
- Sprint planning, documentation, point of contact for designers and technical artists.
- Maintenance of the project's CI/CD pipeline, delivery of new versions and game editor builds.
- Technologies: Snowdrop, C++, **Perforce**, **Jira**, Confluence, Kibana

Game Engine Programmer (Alterra, cancelled project)

Ubisoft

September 2022–January 2024

- Snowdrop game engine and editor tools programming (voxel world, skeletal animation).
- Developing new features for the game editor and debugging existing issues
- Delivering a multi-platform deterministic game experience on PC, Xbox, Playstation using Clang
- Managed the codebase unit test suite using the Catch2 testing framework.
- Administrating local instances of Doxygen and SonarQube to guarantee the quality of the codebase.
- Technologies: Snowdrop, C++, C#, **Perforce**, GitLab, Sharpmake, Clang

Video Game Programming Teacher

ISART Digital, Montréal, QC, Canada

 **ISART**
DIGITAL
September 2019–July 2022

- Teaching and management of game programming classes (1st and 2nd year). My responsibilities included:
- On-site teaching, live coding classes, promotion at industry events, planning and curriculum updates.
- Some of the main subjects I used to teach:
- Basics of C and object-oriented programming with C++
- Advanced C++ (Multithreading, atomics, network sockets)
- 3D rendering, computer graphics and linear algebra (Rasterization, OpenGL, Vulkan, PBR, skeletal animation)
- Pluridisciplinary collab projects in Unity and Unreal Engine with design and art classes
- A Minecraft clone project and a FPS 3D engine, written in C++
- Technologies: C++, C#, OpenGL, Vulkan, PhysX, **Unreal Engine 4**, Unity, **Git**, **Perforce**

Principal Programmer (Front Line Zero, unreleased)

Metatek, London, England (remote)


November 2017–2021

- Freelance work on the Unreal Engine 4 video game project **Front Line Zero**.
- I wrote most of the gameplay code for the game: main character, enemies, bosses, UIs.
- I administrated a Gitea server hosted on a VPS for the team to easily collaborate.
- I helped with the game design and wrote the first draft of the overall plot and some dialogues.
- Technologies: C++, **Unreal Engine 4**, Gitea, Windows, Debian Linux

Game Engine Programmer (The Crew 2)

Ubisoft Ivory Tower, Lyon, France

November 2016–July 2019

- Game engine development, support and tooling for the video game project **The Crew 2**
- Programming and performance profiling on console development kits
- Developed new game engine features
- Optimized the codebase to run on PC, Xbox One, Playstation 4.
- Debugging lock-free utilities in asynchronous resource manager code
- Various editor tools for artists and designers, curve editor and a 2D map editing tool
- Implementing a time drift compensation algorithm, server and client-side
- Prototype of boids simulation for flying wildlife (birds)
- Distance-based entity spawning refactoring (mesh instancing optimization)
- Technologies: **C++**, CMake, Perforce, Windows, Xbox One, Playstation 4



September 2014–August 2016

Tools Programmer (Dishonored 2)

Arkane Studios, Lyon, France

- Maintenance and development of tools for the video game project **Dishonored 2**
- Developed an automated texel and vertex density checking tools
- Refactored the game editor's procedural eye animation tool
- Developed a pose exporter for in-editor NPC previewing
- SQL Server database administration and development of a companion REST web app
- Technologies: **C++**, Python, C#, SQL Server, AngularJS, Perforce, Windows



May 2013–August 2013

Robotics Developer

Metalab Reticular Art Center, Lyon, France

- Programming of a Magabot robot for remote control over wireless devices
- Supervision of the project's design documents and development of a first prototype
- Technologies: **Arduino C++**, **Python**, Ubuntu Linux, Git



November 2012–April 2013

VTK C++ Programmer

Orten, Lyon, France

- Profiling and performance improvements of the company's medical 3D editor
- Development of an open-source multi-texturing helper for the VTK 3D toolkit
- Technologies: **C++**, Qt, VTK, SVN, Windows 7

Backend Programmer

Hexaglobe, Paris, France

- Development of a website analytics platform tracking advertising revenue and visitors geolocation
- Backend acceleration using Memcache
- Geolocation frontend using highcharts.js
- Technologies: **PHP**, **Symfony**, MySQL, jQuery, Ubuntu Linux, SVN



July 2011–December 2011

Technical skills

Tools: RenderDoc, NSight, CMake, Visual Studio

Platforms: PC (Windows/Linux), Xbox Series X, Playstation 5

Version control systems: Git, Perforce, SVN

Languages: **C++**, **Python**, C#, JavaScript

Rendering: OpenGL, Vulkan, D3D11, Emscripten, GLSL, HLSL

Game Engines: Anvil, Snowdrop, UE5, Unity, Dunia, Void Engine

Education

Master of Information Technology

2010–2015

European Institute of Technology (EPITECH), Lyon, France

Certificate in Information Technology

2013–2014 (exchange year)

Université Laval, Québec, Canada

Spoken languages

French (native), English (fluent), Spanish (comprehension), Persian (intermediate), Japanese (beginner)